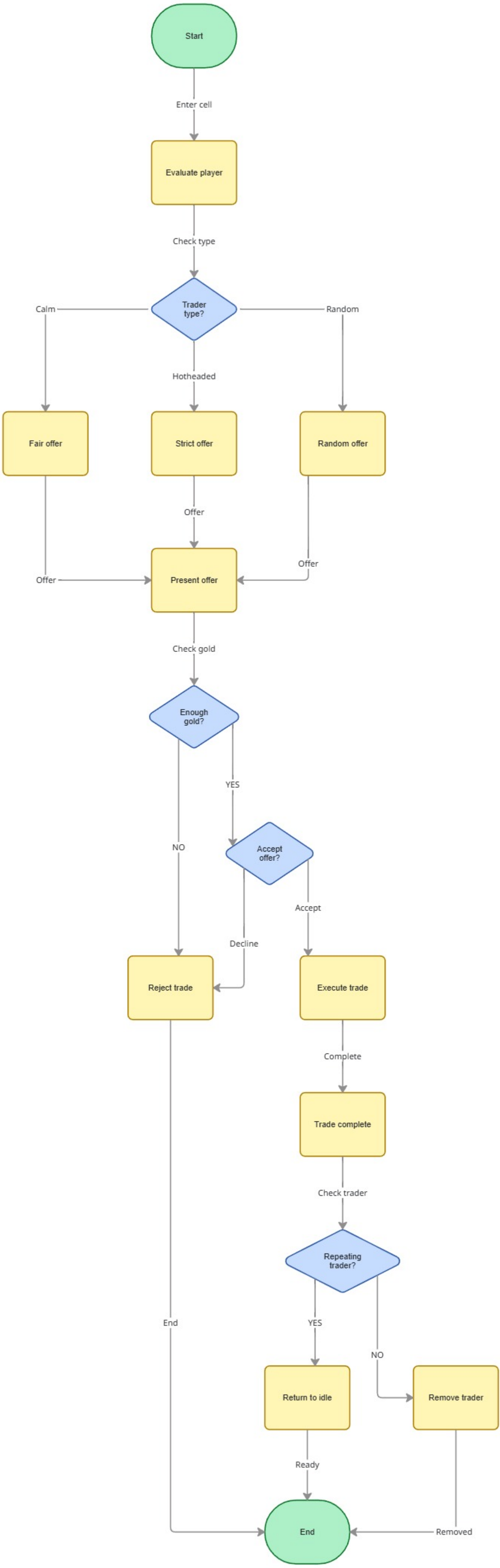
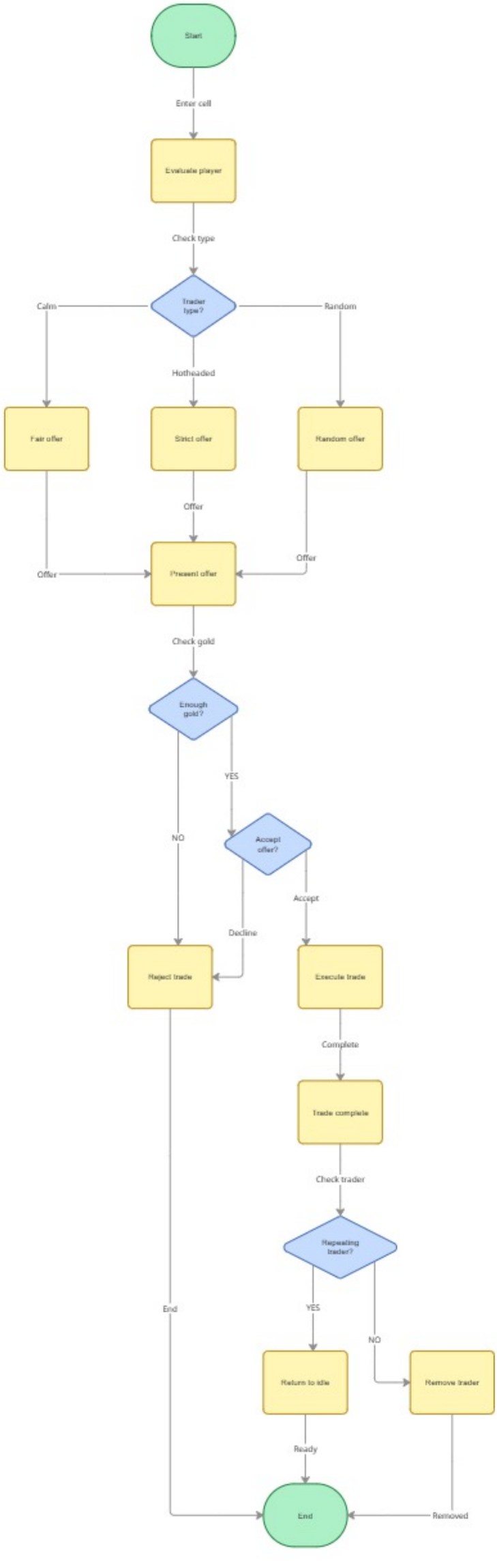

Start
Enter cell
Evaluate player
Check type
Trader type?
Calm
Hotheaded
Random
Fair offer
Strict offer
Random offer
Offer
Offer
Offer
Present offer
Check gold
Enough gold?
YES
NO
Accept offer?
Accept
Decline
Reject trade
Execute trade
Complete
Trade complete
Check trader
Repeating trader?
YES
NO
End
Return to idle
Remove trader
Ready
Removed
End


Start
Enter cell
Evaluate player
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Trader type?
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Random
Fair offer
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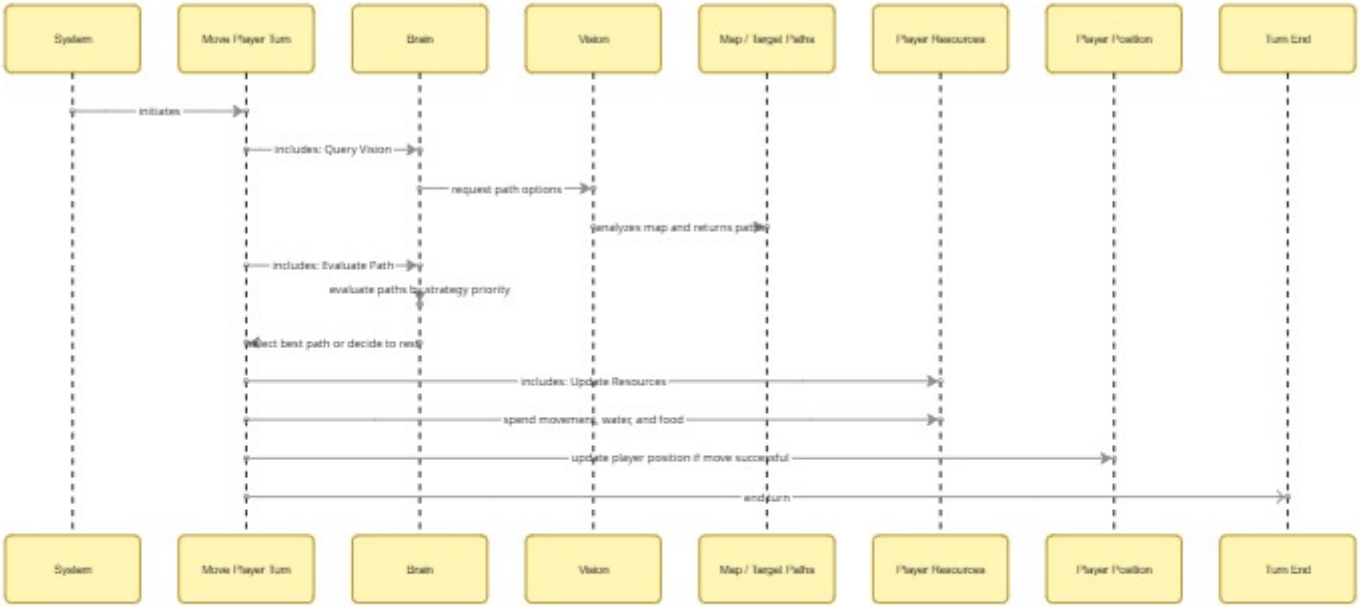
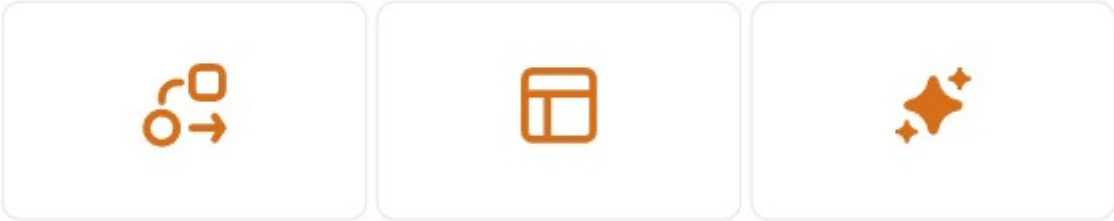
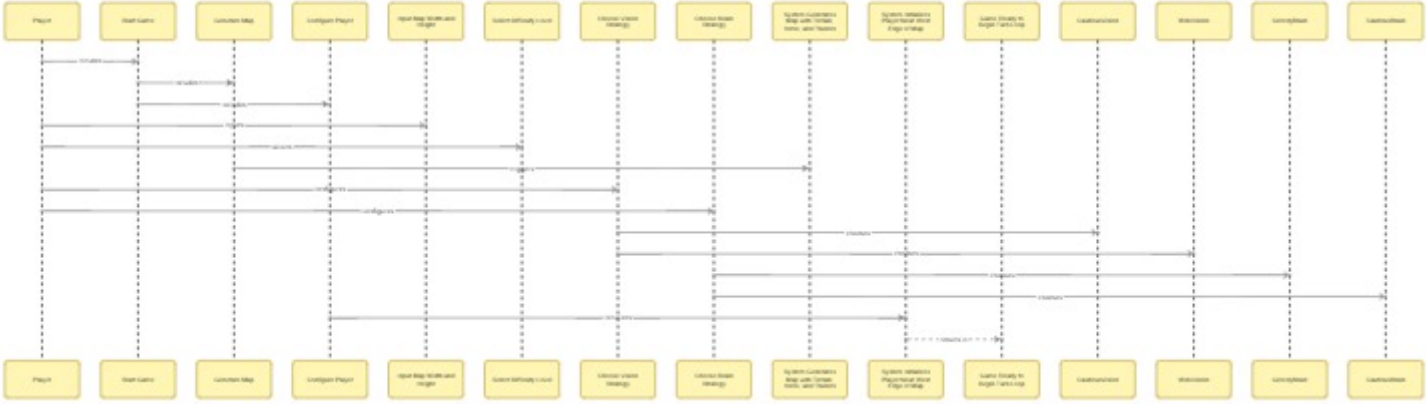
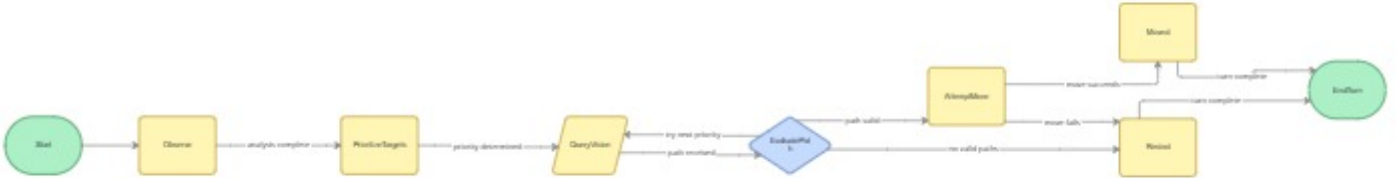
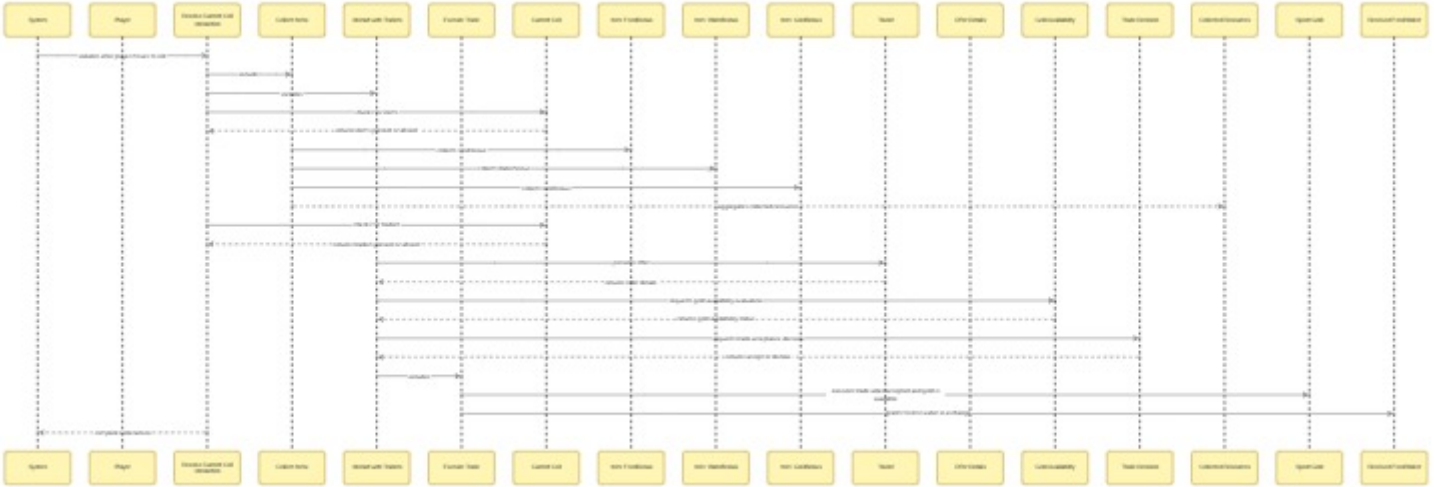
Removed

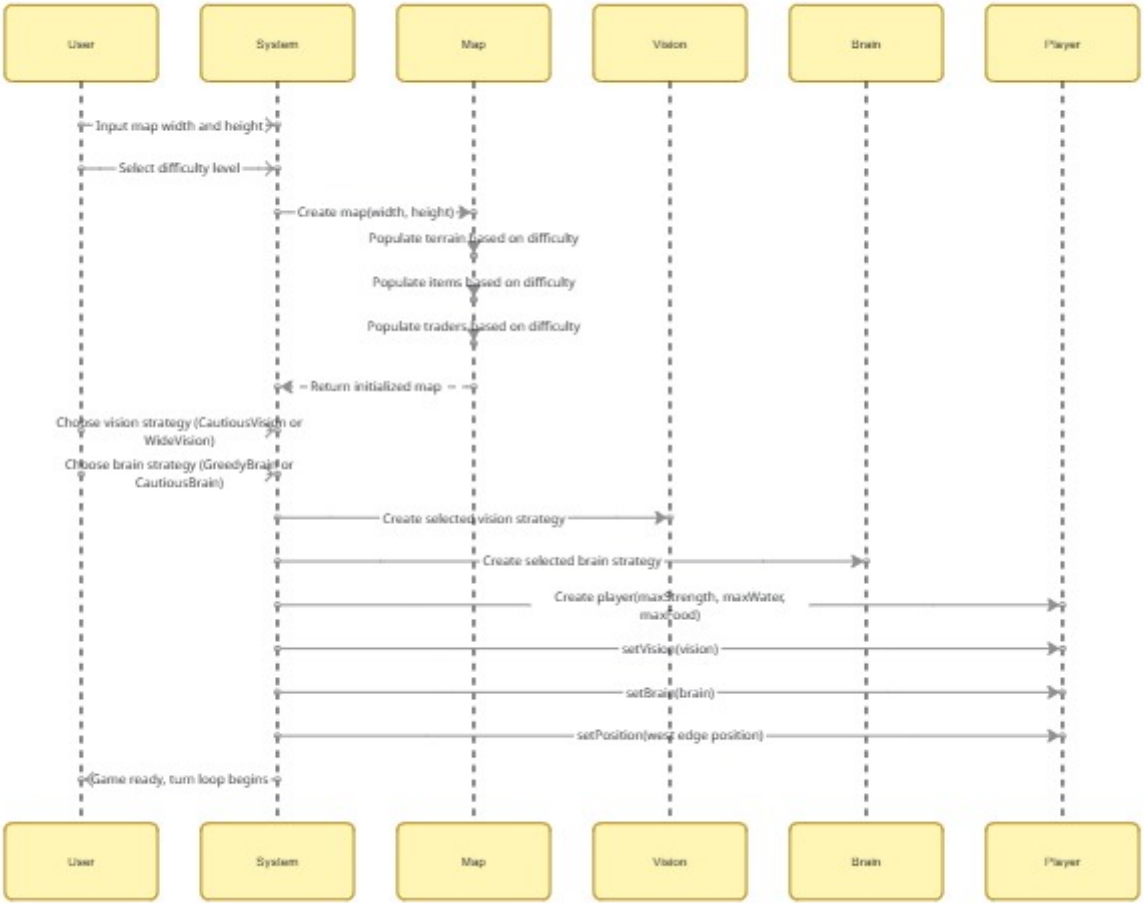

System
Move Player Turn
Brain
Vision
Map / Target Paths
Player Resources
Player Position
Turn End
initiates
includes: Query Vision
request path options
analyzes map and returns paths
includes: Evaluate Path
evaluate paths by strategy priority
select best path or decide to rest
includes: Update Resources
spend movement, water, and food
update player position if move successful
end turn
System
Move Player Turn
Brain
Vision
Map / Target Paths
Player Resources
Player Position
Turn End

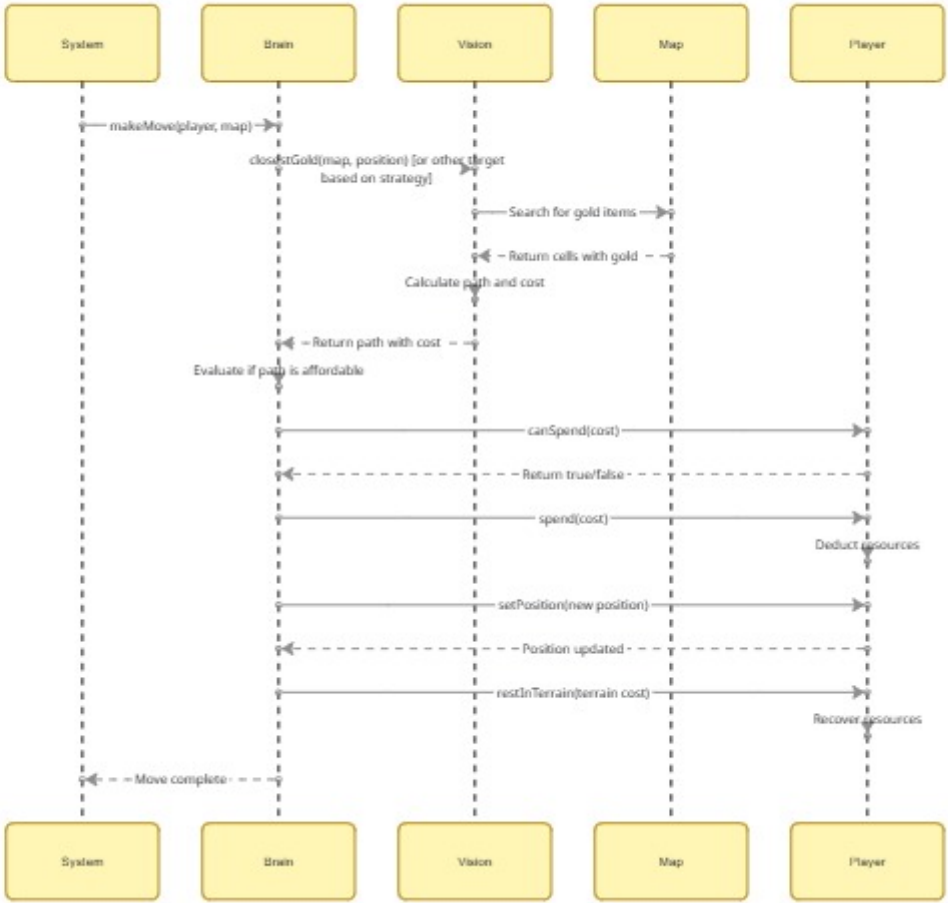


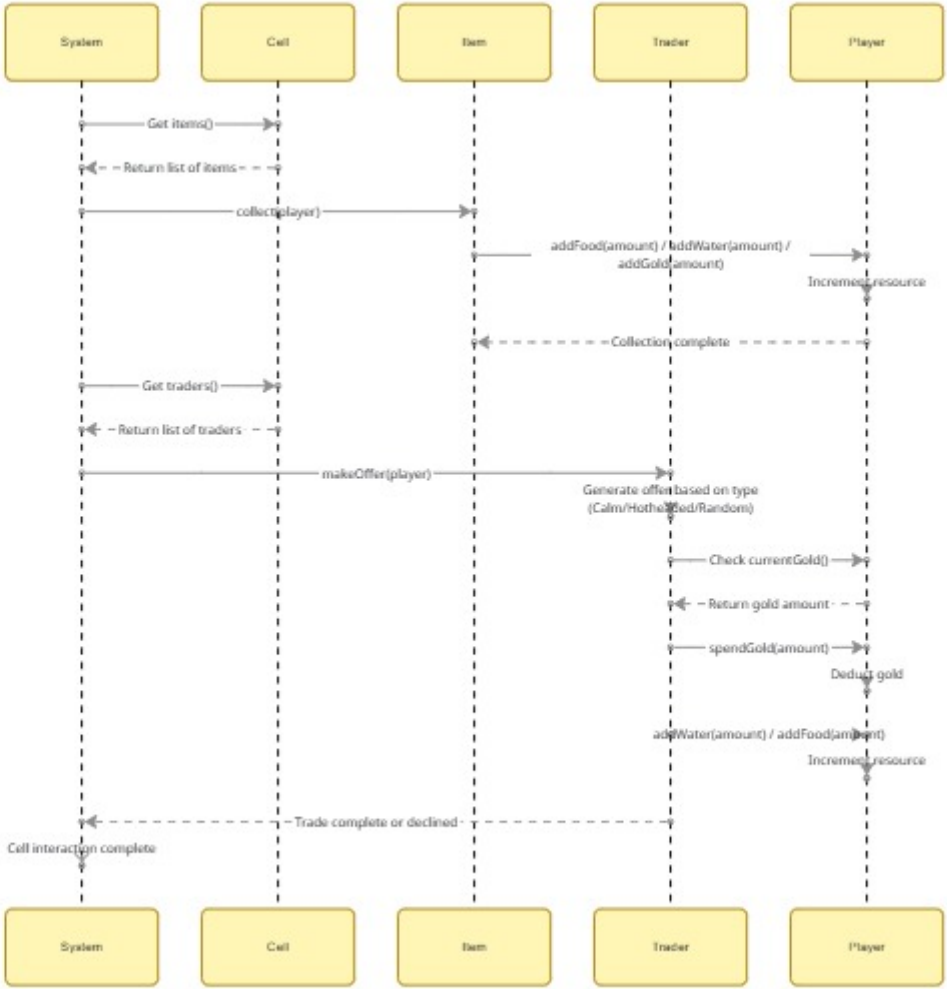


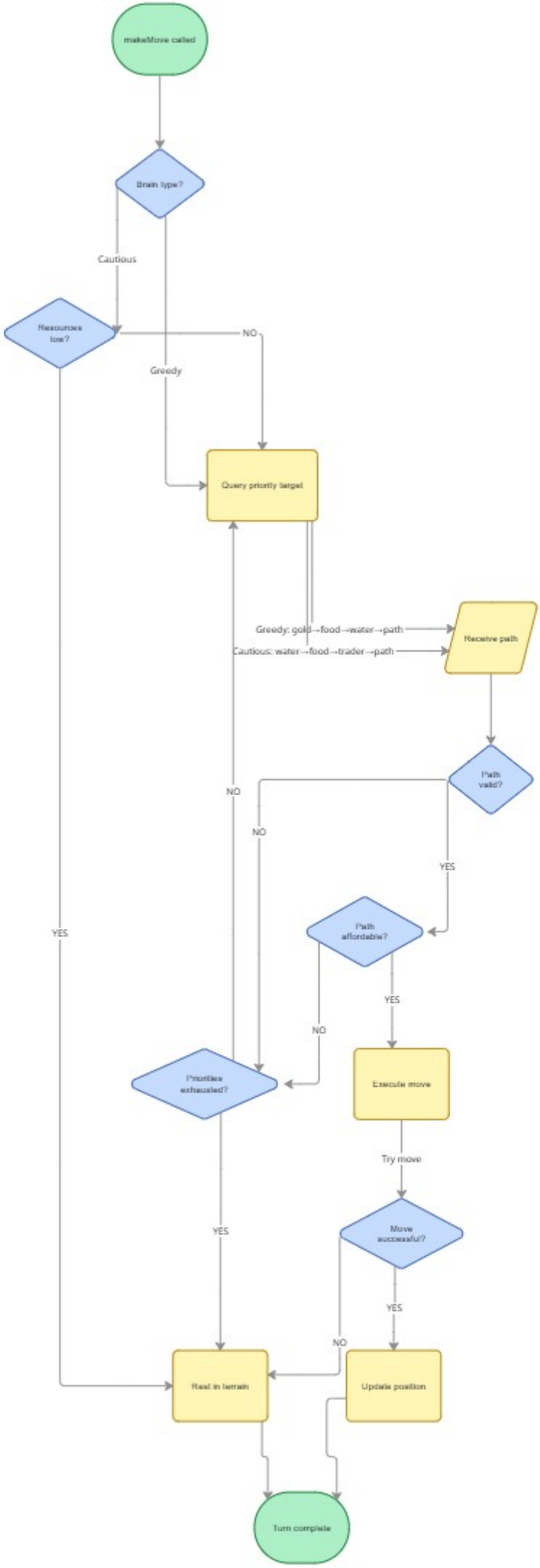

Start
Observe
analysis complete
Prioritize/Targets
priority determined
QueryVision
try next priority
path exhausted
EvaluatePath
path valid
no valid paths
AttempMove
move succeeds
move fails
Moved
Revised
user complete
user complete
EndTurn




User
System
Map
Vision
Brain
Player
Input map width and height
Select difficulty level
Create map(width, height)
Populate terrain based on difficulty
Populate items based on difficulty
Populate traders based on difficulty
Return initialized map
Choose vision strategy (CautiousVision or WideVision)
Choose brain strategy (GreedyBrain or CautiousBrain)
Create selected vision strategy
Create selected brain strategy
Create player(maxStrength, maxWater, maxFood)
setVision(vision)
setBrain(brain)
setPosition(west edge position)
Game ready, turn loop begins
User
System
Map
Vision
Brain
Player


System
Brain
Vision
Map
Player
makeMove(player, map)
closestGold(map, position) [or other target based on strategy]
Search for gold items
Return cells with gold
Calculate path and cost
Return path with cost
Evaluate if path is affordable
canSpend(cost)
Return true/false
spend(cost)
Deduct resources
setPosition(new position)
Position updated
restInTerrain(terrain cost)
Recover resources
Move complete
System
Brain
Vision
Map
Player


System
Cell
Item
Trader
Player
Get items()
Return list of items
collect(player)
addFood(amount) / addWater(amount) / addGold(amount)
Increment resource
Collection complete
Get traders()
Return list of traders
makeOffer(player)
Generate offer based on type (Calm/Hothead/Random)
Check currentGold()
Return gold amount
spendGold(amount)
Deduct gold
addWater(amount) / addFood(amount)
Increment resource
Trade complete or declined
Cell interaction complete
System
Cell
Item
Trader
Player


makeMove called
Brain type?
Cautious
Greedy
Resources low?
NO
YES
Query priority target
Greedy: gold→food→water→path
Cautious: water→food→trader→path
Receive path
Path valid?
NO
NO
YES
Path affordable?
YES
NO
Priorities exhausted?
Execute move
Try move
Move successful?
YES
YES
NO
Read in terrain
Update position
Turn complete